CS44112: Artificial Intelligence
L-T-P-Cr: 3-0-2-4

Introduction to Artificial Intelligence

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UNIT I: Intelligent agents **Lectures: 5**

Reactive, deliberative, goal-driven, utility-driven, and learning agents. Artificial Intelligence programming techniques.

UNIT II: Problem-solving through Search **Lectures: 5**

Forward and backward, state-space, blind, heuristic, problem-reduction, A, A*, AO*, minimax, constraint propagation, neural, stochastic, and evolutionary search algorithms, sample applications.

UNIT III: Knowledge Representation and Reasoning **Lectures: 7**

ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time, and space; predicate logic, situation calculus, description logics, reasoning with defaults, reasoning about knowledge, sample applications

UNIT IV: Planning **Lectures: 4**

Planning as search, partial order planning, construction and use of planning graphs

UNIT V: Representing and Reasoning with Uncertain Knowledge **Lectures: 5**

Probability, connection to logic, independence, Bayes rule, Bayesian networks, probabilistic inference, sample applications.

UNIT VI: Decision-Making **Lectures: 5**

Basics of utility theory, decision theory, sequential decision problems, elementary game theory, sample applications.

UNIT VII: Machine Learning and Knowledge Acquisition **Lectures: 10**

Learning from memorization, examples, explanation, and exploration learning nearest neighbour, naive Bayes, and decision tree classifiers, Q-learning for learning action policies, applications. Sample Applications of AI, student project presentations.

UNIT VIII: Brief Survey of selected additional topics **Lectures: 3**

Perception, communication, interaction, and action; multi-agent systems.

The goal for this course:

- To introduce you to the field of Artificial Intelligence.
- To explain the challenges inherent in building an “intelligent system”
- To explain the
  - Key Paradigms
  - Core Techniques
  - Algorithms for some of the techniques

Instructional Objectives

On taking this course you should be able to:

- Understand the role of basic
  - Formulate problems as state space search problems and efficiently solve them
  - Write game playing problems
  - Use machine learning to find patterns
  - Build expert systems
  - Knowledge representation,
  - Problem solving, and Learning methods in AI : In engineering intelligent systems
  - You will be able to assess the applicability, strengths, and weakness of methods in solving particular engineering problems. While looking to different components of intelligent…
  - You will learn, how to develop intelligent systems by assembling solutions to concrete computational problems.

Textbooks:

1) *Artificial intelligence and intelligent systems* by N. P. Padhy, Published by Oxford University Press.
2) *Artificial Intelligence: A Modern Approach*, 3rd Edition, by Stuart Russell and Peter Norvig.
3) *Machine Learning*, Tom Mitchell, McGraw Hill, 1997.

Grading (Theory)

Midterm	**(30%)**
Class test/Assignment	**(10%)**
Participation	**(5% included in Assignment)**
Final	**(60%)**

List of lab assignments:

Assignment:

- **Assignment is very important. It is the best way for you to learn the material. You are encouraged to discuss the problems with your classmates,** *but all work handed in* ***should be original, and written by you in your own words****.*

AI: Goals

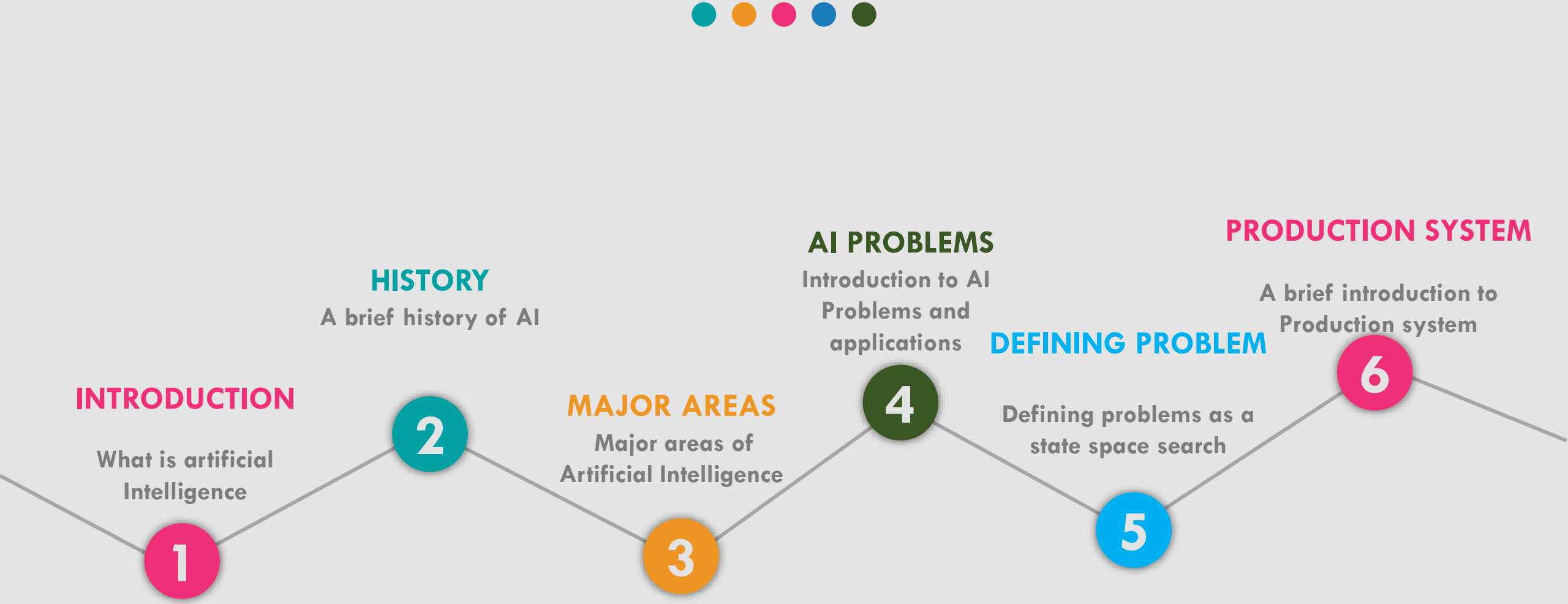
- **Ambitious goals:**
  - **understand “intelligent” behavior**
  - **build “intelligent” agents**

What is Intelligence?

- **Intelligence:**
  - **capacity to learn and solve problems**
  - **the ability to act rationally-** means thinking and making decisions based on clear reason or logic

"A rational agent is someone who uses intelligent thinking instated of emotions when making choices"

TOPICS


INTRODUCTION
What is artificial Intelligence
1
HISTORY
A brief history of AI
2
MAJOR AREAS
Major areas of Artificial Intelligence
3
AI PROBLEMS
Introduction to AI Problems and applications
4
DEFINING PROBLEM
Defining problems as a state space search
5
PRODUCTION SYSTEM
A brief introduction to Production system
6

TOPICS

WHY ?

Why we are learning AI

7

WHERE ?

Where it will help us

8

HOW?

Possible ways to dig into the AI

9

QUERIES

Clear you mind by asking Intelligent questions

10

INTRODUCTION

What is artificial Intelligence

1

INTRODUCTION

- What is AI?

Views of AI fall into four different perspectives
--- two dimensions:

1) Thinking versus Acting
2) Human versus Rational (which is "easier"?)



AI refers to the ability of machines to perform cognitive tasks like thinking, perceiving (become aware of through the senses), learning, problem-solving and decision making. Initially believed as a technology that could mimic human intelligence.

With incredible advances made in data collection, processing and computation power, intelligent systems can now be deployed to take over a variety of tasks, enable connectivity and enhance productivity.

Rational behavior refers to a decision-making process that is based on making choices that result in the optimal level of benefit or utility for an individual.

INTRODUCTION

- Definition – According to research of *Stuart J. Russell and Peter Norvig*
- It is divided into 4 parts (categories)

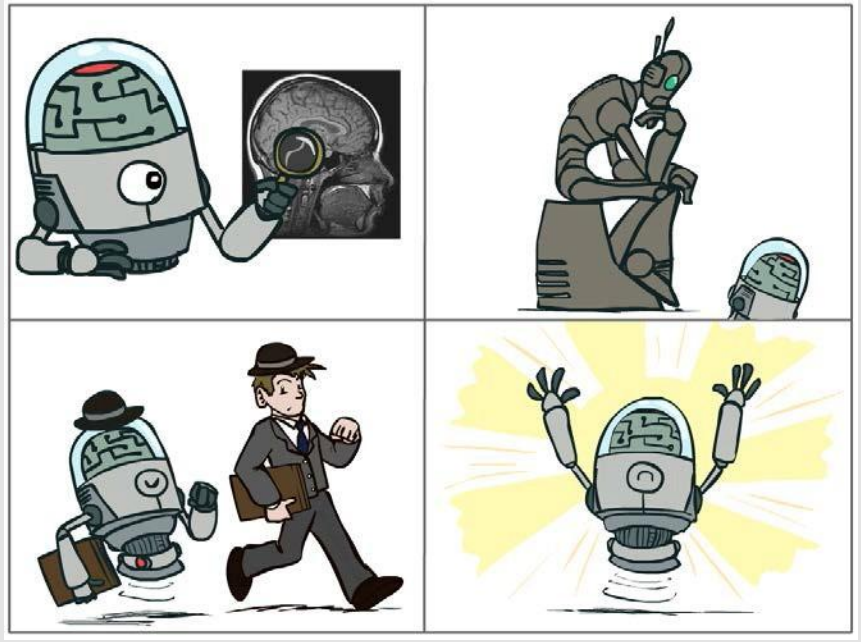
Systems that think like humans.	Systems that think rationally.
Systems that act like humans.	Systems that act rationally.

AI is about building rational agents.
An agent is something that perceives and act.

Approaches to AI

The science of making machines that:

Think like people



Think rationally

Act like people

Act rationally

Computers/machines require external assistance—they do not think by themselves.

Some definitions of AI organized into four categories

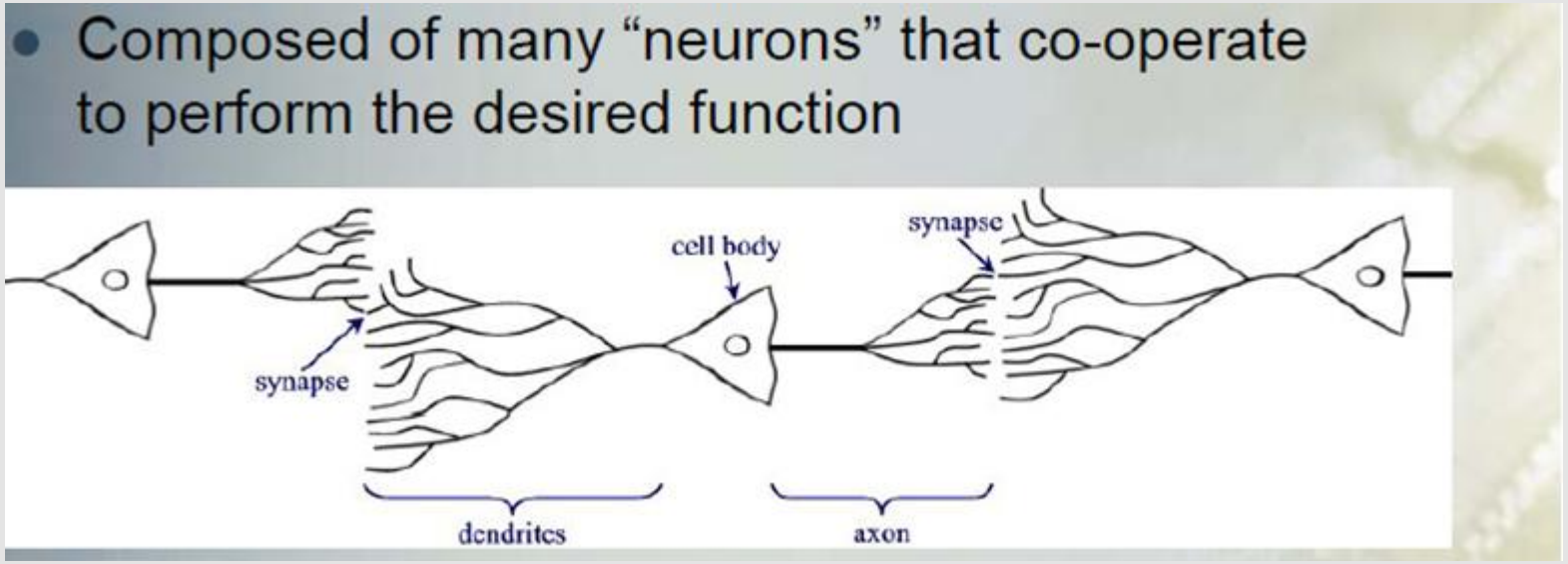
	LEFT: Fidelity to **human** performance	RIGHT: Ideal performance measure, called ***rationality***
TOP: **Thought** processes and **Reasoning**	Thinking Humanly • "The exciting new effort to make computers think ...machines with minds, in the full and literal sense." *(Haugeland, 1985)* • "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ..." *(Bellman, 1978)*	Thinking Rationally • "The study of mental faculties through the use of computational models." *(Charniak and McDermott, 1985)* • "The study of the computations that make it possible to perceive, reason, and act." *(Winston, 1992)*
BOTTOM: **Behavior**	Acting Humanly • "The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) • "The study of how to make computers do things at which, at the moment, people are better." *(Rich and Knight, 1991)*	Acting Rationally • "Computational Intelligence is the study of the design of intelligent agents." *(Poole et al., 1998)* • "AI ...is concerned with intelligent behavior in artifacts." *(Nilsson, 1998)*

Human intelligence

- In general there are various reasons why trying to mimic humans might **not** be the best approach to AI:
  - Computers and Humans have a very different architecture with quite different abilities.
    - Numerical computations
    - Visual and sensory processing
    - Massively and slow parallel vs. fast serial

	Computer	Human Brain
Computational Units	1 CPU, 10^8 gates	10^{11} neurons
Storage Units	10^{11} bits RAM 10^{12} bits disk	10^{11} neurons 10^{14} synapses
Cycle time	10^{-9} sec	10^{-3} sec
Bandwidth	10^{10} bits/sec	10^{14} bits/sec
Memory updates/sec	10^9	10^{14}

How does the brain work?



In brain a neuron has three principal components:

1. **Dendrites:-** that carry electrical signals into the cell body.
2. **Cell Body:-** effectively sums and thresholds these incoming signals.
3. **Axon:-** is a single long fiber that carries the signal from the cell body out to other neurons.
4. The point of contact between an axon of one cell and a dendrite of another cell is called a '**synapse**'

Rationality

- The alternative approach relies on the notion of **rationality**.
- Typically this is a precise mathematical notion of what it means to *do the right thing* in any particular circumstance. Provides
  - A precise mechanism for analyzing and understanding the properties of ideal behaviour we are trying to achieve.

Rational Agent

- AI is about building rational agents.
- An agent is something that perceives and act.
- A rational agent always does the right thing. Right thing?
  - What are the functionalities (goals)?
  - What are the components?
  - How do we build them?

Rationality

- Perfect Rationality
  - Assumes that the rational agent knows all and will take the action that maximizes her utility.
  - Human beings do not satisfy this definition of rationality. However perfect rationality is something that is not within our scope. Many times we cannot reason in a given time frame.
- Bounded Rationality Herbert Simon, 1972 (of CMU)
  - Because of the limitations of the human mind, humans must use approximate methods to handle many tasks.
  - The bounded rationality does not have the aim to maximize the utility. So it is for agent to take best action by given limited resources.

HISTORY

A brief history of AI

2

A Brief History of AI

Introduction

The intellectual roots of AI, and the concept of intelligent machines, may be found in Greek mythology. Intelligent artifacts appear in literature since then, with real (and fraudulent) mechanical devices actually demonstrated to behave with some degree of intelligence. Some of these conceptual achievements are listed below under "Ancient History."

After modern computers became available, following World War II, it has become possible to create programs that perform difficult intellectual tasks. From these programs, general tools are constructed which have applications in a wide variety of everday problems. Some of these computational milestones are listed below under "Modern History."

Ancient History

Greek myths of Hephaestus, the blacksmith who manufactured mechanical servants, and the bronze man Talos incorporate the idea of intelligent robots. Many other myths in antiquity involve human-like artifacts. Many mechanical toys and models were actually constructed, e.g., by Archytas of Tarentum, Hero, Daedalus and other real persons.

4th century B.C.

- Aristotle invented syllogistic logic, the first formal deductive reasoning system.

20th century - First Half

- Bertrand Russell and Alfred North Whitehead published *Principia Mathematica*, which revolutionaized formal logic. Russell, Ludwig Wittgenstein, and Rudolf Carnap lead philosophy into logical analysis of knowledge.
- Torres y Quevedo built his chess machine 'Ajedrecista', using electromagnets under the board to play the endgame rook and king against the lone king, possibly the first computer game (1912).

Modern History

The modern history of AI begins with the development of stored-program electronic computers. For a short summary, see Genius and Tragedy at Dawn of Computer Age By ALICE RAWSTHORN, NY Times (March 25, 2012), a review of technology historian George Dyson's book "Turing's Cathedral: The Origins of the Digital Universe."

1956

The A.I. TIMELINE

Ramon Llull

-Catalan poet and theologian.

"The Ultimate General Art"

Create new knowledge from combinations of concepts.

1308

1666

Gottfried Leibniz publishes

-Mathematician and philosopher

"On the Combinatorial Art"

Combinations of a relatively small number of simple concepts.

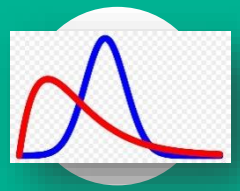
Jonathan Swift



Publishes *Gulliver's Travels*, which includes a description of the Engine, a machine on the island of Laputa (and a parody of Llull's ideas)

1726

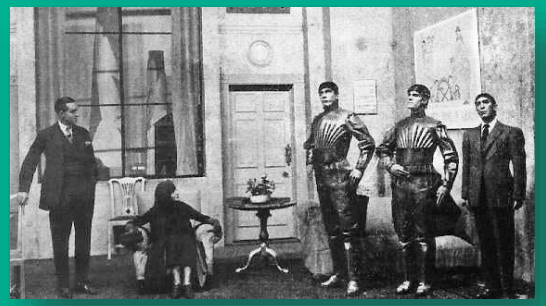
1763

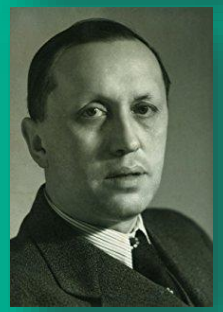


Thomas Bayes

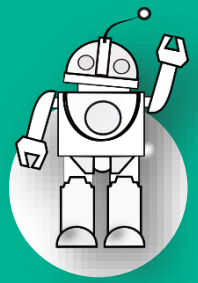


Develops a framework for reasoning about the probability of events.
"Bayesian inference"





Karel Čapek



Introduces the word "robot" in his play R.U.R. (Rossum's Universal Robots).

1921

The "standard interpretation" of the Turing test, in which player C, the interrogator, is given the task of trying to determine which player – A or B – is a computer and which is a human. The interrogator is limited to using the responses to written questions to make the determination.

1950

Alan Turing



Publishes “Computing Machinery and Intelligence” in which he proposes “the imitation game” which will later become known as the “Turing Test”

1951

Marvin Minsky

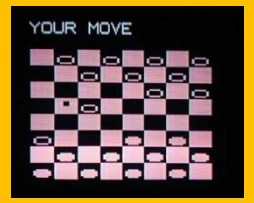


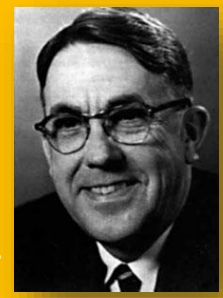


This a sample text and here you need to put your own text and ideas that fit. ***“First ANN”***, using 3000 vacuum tubes to simulate a network of 40 neurons

1952

Arthur Samuel





Develops the **“first computer checkers-playing program”** and the first computer program to learn on its own.

Term “artificial intelligence” is coined

John McCarthy (Dartmouth College),
Marvin Minsky (Harvard University),
Nathaniel Rochester (IBM),
and Claude Shannon (Bell Telephone Laboratories)

1955

1961



General Motors

The first industrial robot, *“Unimate”*, starts working on an assembly line in a General Motors plant in New Jersey.

ELIZA

An interactive program that carries on a dialogue in English language on any topic.



1965

1st Expert System

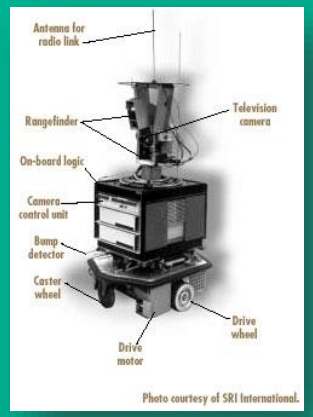
Edward Feigenbaum, Bruce G. Buchanan, Joshua Lederberg, and Carl Djerassi start working on **DENDRAL** at Stanford University.

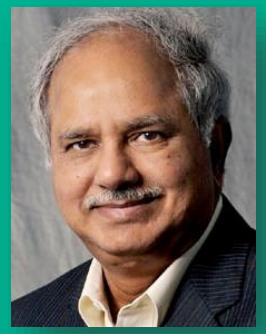
1965

1966

SHAKEY

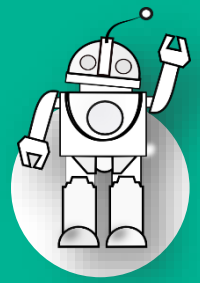
First general-purpose mobile robot to be able to reason about its own actions.





Raj Reddy

"Speech Recognition by Machine: A Review" in the Proceedings of the IEEE, summarizing the early work on Natural Language Processing (NLP).



1976

1979

The Stanford Cart

One of the earliest examples of an autonomous vehicle.





A.L.I.C.E

(Artificial Linguistic Internet Computer Entity)
The chatbot-inspired by Joseph Weizenbaum's ELIZA program

1995



1997

Deep Blue

Becomes the first computer chess-playing program to beat world chess champion.



A driverless car

Google starts developing. In 2014, it became the first to pass, in Nevada, a U.S. state self-driving test.

2009

2011

IBM's Watson

A natural language question answering computer.



AlphaGO

Google DeepMind's AlphaGo defeats Go champion Lee Sedol.

2016



10 Major MILESTONES

HISTORY in SHORT ….!!!

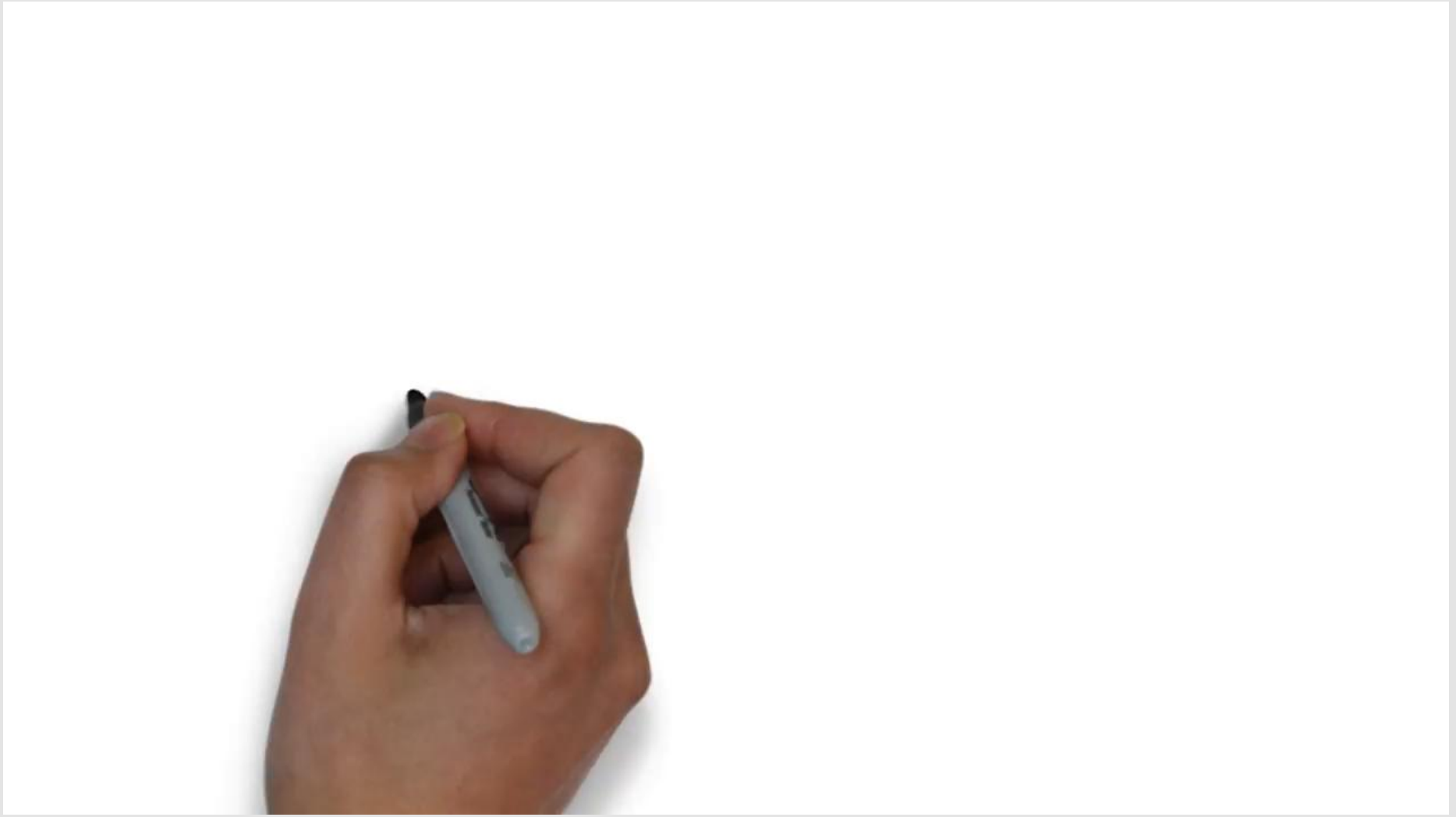


MAJOR AREAS

Major areas of Artificial Intelligence



Major areas of AI



Key research areas in AI

- **Problem solving, planning, and search --- generic problem solving architecture based on ideas from cognitive science (game playing, robotics).**
- **Knowledge Representation – to store and manipulate information (logical and probabilistic representations)**
- **Automated reasoning / Inference – to use the stored information to answer questions and draw new conclusions**
- **Machine Learning – intelligence from data; to adapt to new circumstances and to detect and extrapolate patterns**

Cognitive science is the study of the human mind and brain, focusing on how the mind represents and manipulates knowledge and how mental representations and processes are realized in the brain.

Key research areas in AI (contd…)

- **Natural Language Processing – is a subset of AI. It is related to the interaction between computers and humans using the natural language. (to communicate with the machine)**
  - **The most important objective of NLP is to read, understand, and make sense of human languages in a useful way.**
  - **Computer can easy understand the data that is available in structured forms, like spreadsheets and databases. However, human languages, text, and voices are unstructured forms of data. It is difficult for computer to understand such unstructured data. NLP comes handy in this situation.**
  - **Applications:**
    - **Language translation applications, like Google translate**
    - **Word processors, like MS Word and Grammarly to check grammatical accuracy of texts**
    - **Interactive voice response (IVR) applications to respond to certain users's requests**
    - **Personal assistant applications, line OK Google. Siri, Cortana, and Alexa**

Key research areas in AI (contd…)

- **Computer Vision --- processing visual information**
  - **Applications:**
    - **Healthcare: to examine scan reports and identify patterns. Helping doctors in treating their patients and increasing the survival rates of patients with fatal conditions.**
    - **Banking: Video KYC**
    - **Self-driven cars: A self-driven car is usually fitted with surround cameras along with a dozen sensors to train the car to see, identify obstacles, and read traffic lights etc…**
- **Robotics --- Autonomy, manipulation, full integration of AI capabilities**

Building Intelligent Machines

- **Building exact models of human cognition view from psychology, cognitive science, and neuroscience**

- **Developing methods to match or exceed human performance in certain domains, possibly by very different means**

AI PROBLEMS

Introduction to AI

Problems and applications



It's not that I'm so smart, it's just that I stay with problems longer.

—Albert Einstein
(1879 –1955), German-born theoretical physicist

IDENTIFYING PROBLEMS SUITABLE FOR AI

“What makes a problem suitable for AI?”

- There are three characteristics that are common to most AI problems:
  - AI problems tend to be large.
  - They are computationally complex and cannot be solved by straightforward algorithms.
  - AI problems and their domains tend to embody a large amount of human expertise, especially if tackled by strong AI methods.
- For example
  - Medical diagnosis
  - ATMs
  - Two person games such as chess and checkers

DEFINING PROBLEM

Defining problems as a state space search



Defining the Problems

- **To build a system to solve a particular problem, we need to do four things:**

1. Define the problem precisely. This definition must include precise specifications of what the initial situation (s) will be as well as what final situations constitute acceptable solutions to the problem.
2. Analyze the problem. A few very important features can have an immense impact on the appropriateness of various possible techniques for solving the problem.
3. Isolate and represent the task knowledge that is necessary to solve the problem.
4. Choose the best problem-solving technique(s) and apply it (them) to the particular problem.

PROBLEM CHARACTERISTICS

- Is the problem decomposable into a set of (nearly) independent smaller or easier subproblems?
- Can solution steps be ignored or at least undone if they prove unwise?
- Is the problem's universe predictable?
- Is a good solution to the problem obvious without comparison to all other possible solutions?
- Is the desired solution a state of the world or a path to a state?
- Is a large amount of knowledge absolutely required to solve the problem, or is knowledge important only to constrain the search?
- Can a computer that is simply given the problem return the solution, or will the solution of the problem require interaction between the computer and a person?

Defining the Problems as a State Space Search

- **State space search**
  - It is complete set of states including start and goal states, where the answer of the problem is to be searched.

- **Problem**
  - It is the question which is to be solved. For solving the problem it needs to be precisely defined.

  - The definition means, defining the start state, goal state, other valid states and transitions.

Defining the Problems as a State Space Search

- A state space representation allows for the formal definition of a problem which makes the movement from initial state to the goal state quite easily.

- So we can say that various problems like planning, learning, theorem proving etc. are all essentially search problems only.

- **State space search**
  - It is a process used in the field of computer science, including artificial intelligence (AI), in which successive configurations or states of an instance are considered, with the goal of finding a *goal state* with a desired property.

PROBLEMS

- CHESS:
- WATER JUG
- 8-PUZZLE
- TRAVELLING SALESMAN (Animation: go) "Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?"
- MISSIONARIES AND CANNIBALS (Animation or game: play)
- TOWER OF HANOI (Animation: go)
- MONKEY AND BANANAS

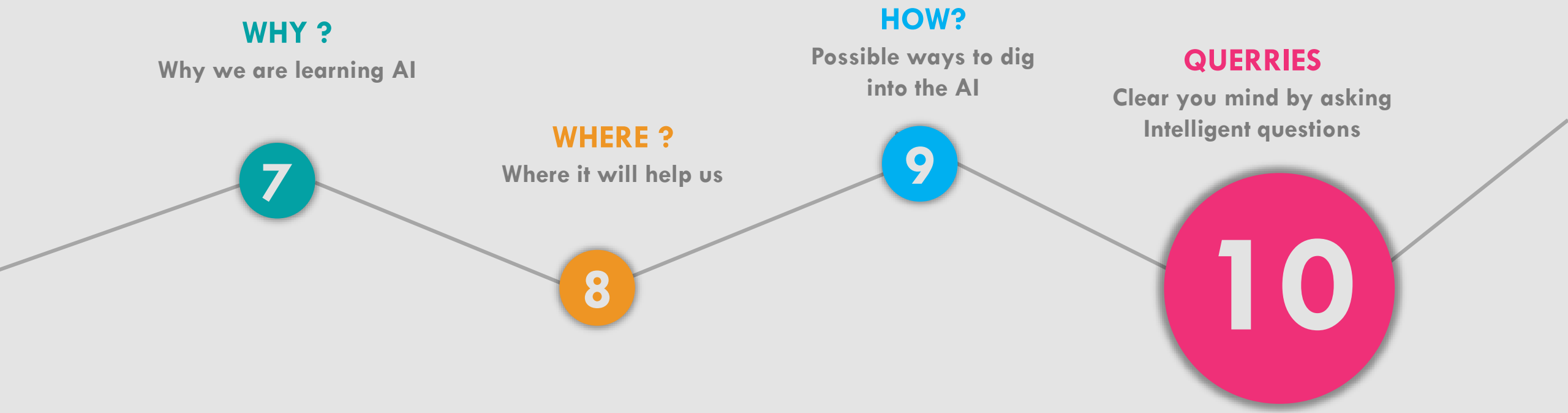
PRODUCTION SYSTEM

A brief introduction to Production system

6

Production Rules and Expert Systems

- Production rules are used in AI as a method of knowledge representation. A production rule has the general forms:
  - IF (Condition), THEN Action
  - or
  - IF (Condition), THEN Fact
- Common examples are:
  - IF (headache), THEN take two aspirins and call me in the morning.
  - IF ((A > B) and (B > C)), THEN A > C


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